



B	A	sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

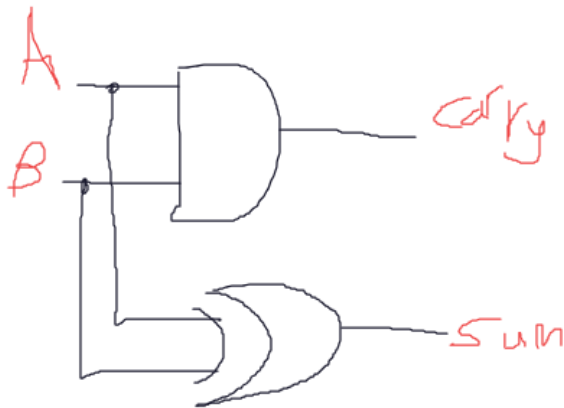
```
//Structural
module FA_S(
    input A, B, Cin,
    output Cout, S
);

    wire sum1, carry1, sum2, carry2;

    HA block1 (A ,B , sum1, carry1);
    HA block2 (sum1, Cin, sum2, carry2);

    assign S= sum2;
    assign Cout= carry1 | carry2 ;

endmodule
```



```
module HA(
    input A, B,
    output sum, carry
);

    assign sum=A^B;
    assign carry=A&B;

endmodule
```

